ICAD RT-5 · Set 10 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

Physics

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
5	a	b	c	d
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18	a	b	c	d
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Numerical — write the value

21	
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Chemistry

1	a	b	c	d
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3	a	b	c	d
4	a	b	c	d
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19	a	b	c	d
20	a	b	c	d

Numerical — write the value

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Mathematics

1	a	b	c	d
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3	a	b	c	d
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19	a	b	c	d
20	a	b	c	d

Numerical — write the value

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Bubbles — single correct · Box — write the value · each subject out of 100

ICAD RT-5 · Set 10

Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

Physics

PART A · Q1–20 · SINGLE CORRECT

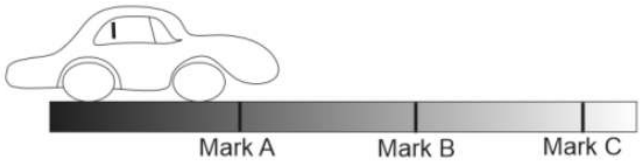
1. P2L2VSuggQ4 If $y = \sin x + 8\cos x + 2\tan x$, then dy/dx is (a) $\cos x - \sin x + 2\sec^2 x$ (b) $\sin x - \cos x + 2\tan^2 x$ (c) $\cos x - 8\sin x + 2\sec^2 x$ (d) $\cos x + 8\sin x + 2\sec x$

2. P4L5V5Q4 The position x of a particle is $x = 9t^2 - t^3$ (metres, seconds). Find the position when the particle achieves maximum speed along +X: (a) 54 m (b) 81 m (c) 24 m (d) 32 m

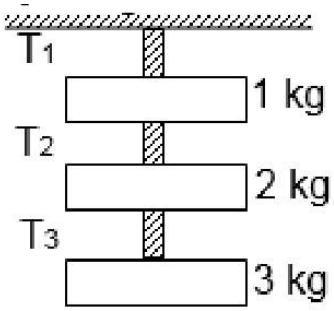
3. P4L8V5Q10 A cart is moving horizontally along a straight line with constant speed 30 m/s. A projectile is to be fired from the moving cart in such a way that it will return to the cart after the cart has moved 80 m. At what speed (relative to the cart) must the projectile be fired? (Take $g = 10\ m/s^2$) (a) 10 m/s (b) $10\sqrt{8}$ m/s (c) 40/3 m/s (d) none

4. P4L8VSuggQ5 A stone projected with a velocity a at an angle θ with the horizontal reaches maximum height H_1. When it is projected with velocity a at an angle $(\pi/2 - \theta)$ with the horizontal, it reaches a maximum height H_2. The relation between the horizontal range R of the projectile, H_1 and H_2 is: (a) $R = 4\sqrt{(H_1H_2)}$ (b) $R = 4(H_1 - H_2)$ (c) $R = 4(H_1 + H_2)$ (d) $R = H_1^2/H_2$

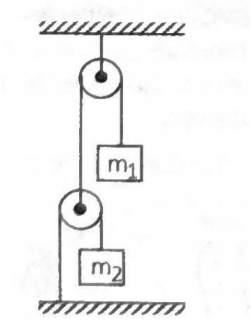
5. P4XVRT2Q18 When a car passes mark-A, driver applies brakes. Thereafter reducing speed uniformly from 160 km/h at A, the car passes mark C with a speed 40 km/h. The marks are at equal distance on the road as shown below. Where on the road was the car moving with a speed 100 km/h? Neglect the size of the car as compared to the distance involved. (a) At mark-B (b) Between mark-A and mark-B (c) Between mark-B and mark-C (d) Information is insufficient to decide.



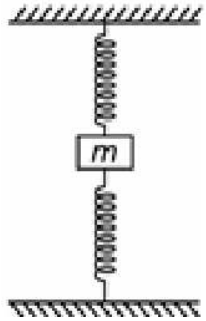
6. P5XVRT4Q1 Find the tension T_2 in the system shown in figure. (a) 1g N (b) 2g N (c) 5g N (d) 6g N



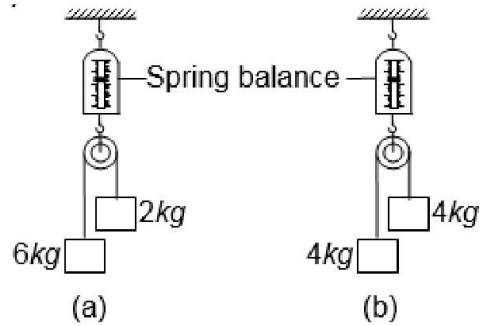
7. P6L3V5Q8 In the system shown, m_1 hangs from one end of a string passing over a fixed pulley; the other end of that string carries a light movable pulley. A second string, tied to the floor, passes over the movable pulley and carries m_2 at its other end. Pulleys are massless and frictionless and the strings are inextensible. The tension in the string connected to m_2 equals (a) $3m_1m_2g/(m_1+4m_2)$ (b) $m_1m_2g/(m_1+4m_2)$ (c) $4m_1m_2g/(m_1+4m_2)$ (d) $5m_1m_2g/(m_1+4m_2)$



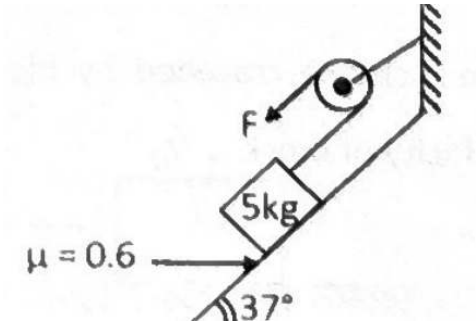
8. P6XVCAT5Q10 A block tied between two springs is in equilibrium. If upper spring is cut, then the acceleration of the block just after cut is 6 m/s² downwards. Now, if instead of upper spring, lower spring is cut, then the magnitude of acceleration of the block just after the cut will be (Take $g = 10\ m/s^2$) (a) 16 m/s² (b) 4 m/s² (c) Cannot be determined (d) None of the above



9. P6XVRT3Q18 Referring to figure (a) and (b) (a) reading of spring balance in (a) is 8 g and in (b) is also 8 g (b) reading of spring balance in (a) is 8 g and in (b), it is less than 8 g (c) reading in (a) is less than 8 g and in (b), it is 8 g (d) reading in both (a) and (b) is less than 8 g



10. P7L2V5Q7 A 5 kg block rests on a rough incline of 37° (μ = 0.6). A force F = 50 N pulls it up the incline through a string passing over a pulley at the top. Find the friction force and its direction on the block by the inclined plane. (g = 10 m/s²) (a) 30 N, up the incline (b) 24 N, down the incline (c) 20 N, up the incline (d) none of these

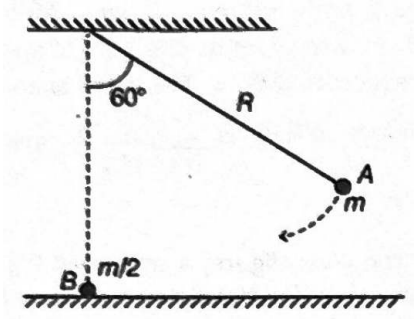


11. P8L2V4Q8 A particle of mass 'm' moves on a straight line with its velocity varying with the distance travelled according to the equation v = a√x, where 'a' is a constant. Find the total workdone by all the forces during a displacement from x = 0 to x = d? (a) mad/2 (b) mad²/2 (c) mad²/4 (d) ma²d/2

12. P8L5V5Q6 The potential energy when a particle moves along the x axis is given by U(x) = 8x² + 2x⁴, where U is in joules and x is the coordinate of the particle in metres. The object exerting the force on the particle is very massive. If the total mechanical energy is 9.0 J, the limits of motion are: (a) −0.96 m, +0.96 m (b) −2.2 m, +2.2 m (c) −1.6 m, +1.6 m (d) −0.96 m, +2.2 m

13. P9L2V5Q8 An object of mass m_1 collides with another object of mass m_2, which is at rest. After the collision, the objects move with equal speeds in opposite direction. The ratio of the masses $m_2 : m_1$ is (a) 3 : 1 (b) 2 : 1 (c) 1 : 2 (d) 1 : 1 (Assume the collision is elastic.)

14. P9L2V6Q3 As shown below, bob A of a pendulum having massless string of length R is released from 60° to the vertical. It hits another bob B of half the mass that is at rest on a frictionless table in the centre. Assuming elastic collision, the magnitude of the velocity of bob A after the collision will be (take g as acceleration due to gravity) (a) √(Rg) (b) (1/3)√(Rg) (c) (2/3)√(Rg) (d) (4/3)√(Rg)



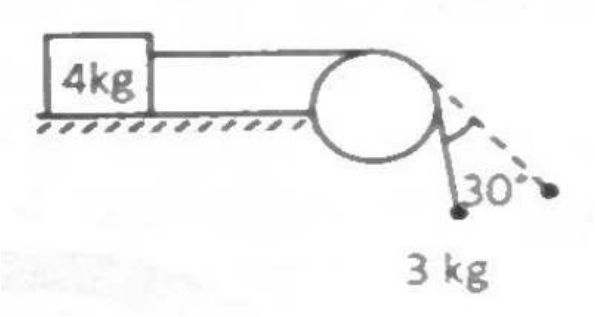
PART B · Q21–25 · NUMERICAL VALUE

21. P3SBVPYQQ3 Resultant of $\bar{A}$ and $\bar{B}$ is perpendicular to $\bar{A}$ and its magnitude is half that of $\bar{B}$ — the angle between $\bar{A}$ and $\bar{B}$ is ___°

22. P4L5VSuggQ5 The displacement of a point moving in a straight line is s = 8t² + 3t − 5 (s in cm, t in s). Find the initial velocity of the particle

15. P10L3V4Q10 A car moves on a circular track of radius r = 100 m. At t = 0, its angular speed is ω 0 = 0.05 rad/s. It speeds up with a constant angular acceleration α = 0.02 rad/s². Find the magnitude of the total acceleration at time t = 10 s. (a) 5.8 m/s² (b) 6.3 m/s² (c) 6.6 m/s² (d) 7.4 m/s²

16. P10L6L7V5Q5 In the system shown mass 3 kg oscillates in a circular arc of amplitude 30°; the minimum coefficient of friction between mass 4 kg and surface of the table to avoid slipping is: (a) 0.25 (b) 0.95 (c) 0.40 (d) 0.60



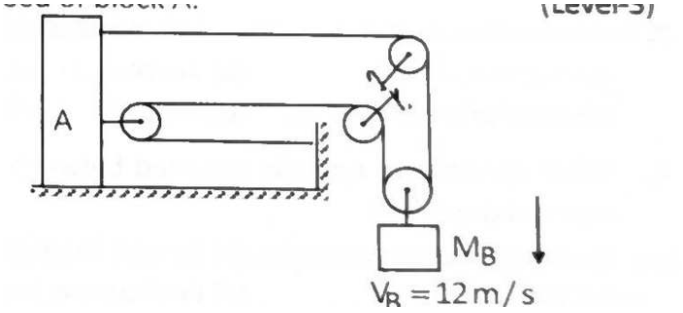
17. P10SBVPYQQ18 A boy ties a stone of mass 100 g to the end of a 2 m long string and whirls it around in a horizontal plane. The string can withstand a maximum tension of 80 N. If the maximum speed with which the stone can revolve is (K/π) rev min −1 , the value of K is: (Assume the string is massless and un-stretchable) (a) 400 (b) 300 (c) 600 (d) 800

18. P10SBVPYQQ19 A stone of mass m, tied to a string, is being whirled in a vertical circle with a uniform speed. The tension in the string is: (a) the same throughout the motion (b) minimum when the rope is in the horizontal position (c) minimum at the highest position of the circular path (d) minimum at the lowest position of the circular path

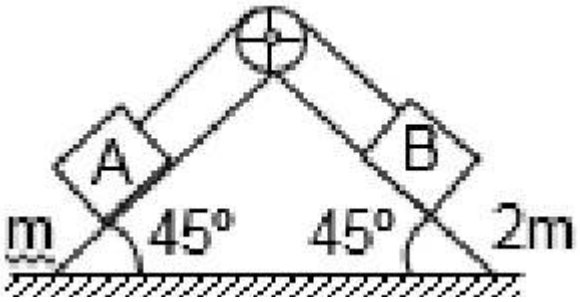
19. P11L3V2Q5 If original area = 12 m 2 and COM is at 6 m, and a part of 3 m 2 is removed from 4 m, find new COM. (a) 6.67 m (b) 6.5 (c) 6.0 m (d) 5.25 m

20. P11L4V2Q5 If two balls each of mass 0.06 kg moving in opposite directions with speed 4 m/s collide and rebound with the same speed, then the impulse imparted to each ball due to other is (a) 0.48 kg.ms (b) 0.24 kg.ms (c) 0.81 kg.ms (d) Zero

23. P6L4VSuggQ3 Find the speed of block A. (One light string: tied to the top of block A, runs right over a fixed pulley, down round the movable pulley carrying block B, up over a second fixed pulley, left round a pulley attached to A, and right to a fixed wall. Block B moves down with V_B = 12 m/s.)



24. P7XVRT3Q22 Block A of mass m and block B of mass 2m are placed on a fixed triangular wedge by means of a massless, inextensible string and a frictionless pulley as shown. The wedge is inclined at 45° to the horizontal on both sides. The coefficient of friction between block A and the wedge is 2/3 and that between block B and the wedge is 1/3. If the blocks A and B are released from rest, the acceleration of A is _____.



25. P11XVCAT6Q22 A uniform thin rod AB of length 'L' has linear mass density μ(x) = a + bx/L where 'x' is measured from 'A'. If the centre of mass of the rod lies at a distance of 7L/12 from 'A' then b/a is

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L4VSuggQ4 A sample of gas is collected over water at 25 °C and barometric pressure 760 mm Hg. Vapour pressure of water at 25 °C = 24 mm Hg. What is the pressure of the dry gas? (a) 736 mm Hg (b) 784 mm Hg (c) 760 mm Hg (d) 24 mm Hg

2. C1SBVPYQQ14 An evacuated glass vessel weighs 40.0 g when empty, 135.0 g when filled with a liquid of density 0.95 g mL^{-1} and 40.5 g when filled with an ideal gas at 0.82 atm at 250 K. The molar mass of the gas in g mol^{-1} is: (Given: R = 0.082 L atm K^{-1} mol^{-1}) (a) 35 (b) 50 (c) 75 (d) 125

3. C2L11VSuggQ4 Which statement regarding 2s and 2p radial probability distributions is correct? (a) Both show one radial node (b) 2p has greater probability near the nucleus than 2s (c) 2s has one radial node whereas 2p has no radial node (d) 2s and 2p have identical angular density

4. C3L1VSuggQ5 Molar mass of electron is nearly: (N_A = 6×10^{23}, m_e = 9.1×10^{-31} kg) (a) 9.1×10^{-31} kg mol^{-1} (b) 9.1×10^{-31} g mol^{-1} (c) 54.6×10^{-8} g mol^{-1} (d) 54.6×10^{-8} kg mol^{-1}

5. C3L11V5Q3 What will be the answer in appropriate significant figures as a result of addition of 3.0223 and 5.041? (a) 80.633 (b) 8.0633 (c) 8.063 (d) 806.33

6. C4L1VSuggQ5 An element has valence shell configuration ns^2 np^4. Identify its period and group based on rules in the theory. (a) Period = n, Group = 14 (b) Period = n, Group = 16 (c) Period = n−1, Group = 6 (d) Period = n+1, Group = 15

7. C4L4V5Q5 Which one of the following elements has the highest ionisation energy? (a) $[Ne]3s^23p^1$ (b) $[Ne]3s^23p^3$ (c) $[Ne]3s^23p^2$ (d) $[Ar]3d^{10}4s^24p^2$

8. C4L6V5Q7 One or more than one option may be correct. Which of the following statement regarding halogens is/are correct? (one or more options may be correct) (a) Ionization energy decreases with increase in atomic number (b) Electronegativity decreases with increase in atomic number (c) Electron affinity decreases with increase in atomic number (d) Enthalpy of fusion increases with increase in atomic number

9. C5L4VSuggQ5 Which pair of species shows the same hybridisation of the central atom, despite having different shapes ? (i) XeF_4 and SF_6 (ii) PCl_5 and XeF_2 (iii) XeF_4 and $[PF_6]^-$ (iv) CO_3^{2-} and BF_3 — (a) (i) only (b) (i) & (iii) (c) (i), (ii) & (iii) (d) All of these

10. C5L5V5Q5 Which of the following two are isostructural? (a) XeF_2, IF_2^- (b) NH_3, BF_3 (c) CO_3^{2-}, SO_3^{2-} (d) PCl_5, ICl_5

11. C5L6VSuggQ5 Consider PH_3, AsH_3, and NH_3. Which statement is correct? (a) Bond angle increases down the group (b) Bond angle decreases $NH_3 \rightarrow PH_3 \rightarrow AsH_3$ due to decreasing EN of central atom (c) All have same bond angle (d) Bond angle depends only on lone pairs

12. C5XVCAT5Q37 If x, y and z represent the bond length of C—H in C_2H_6, C_2H_4 and C_2H_2 respectively then (a) x = y = z (b) x = y > z (c) x < y < z (d) x > y > z

13. C6L1VSuggQ3 Which of the following quantities cannot be used to specify the thermodynamic state of a system? (a) Pressure (b) Heat (c) Temperature (d) Volume

14. C6L7V4Q8 If combustion of 4 g of CH 4 liberates 2.5 kcal of heat, the heat of combustion of CH 4 is: (a) −2 kcal mol −1 (b) −10 kcal mol −1 (c) 2.5 kcal mol −1 (d) −5 kcal mol −1

15. C6L7V4Q9 (ΔH − ΔU) for the formation of carbon monoxide (CO) from its elements at 298 K is: (R = 8.314 J K −1 mol −1) (a) −1238.78 J mol −1 (b) 1238.78 J mol −1 (c) −2477.57 J mol −1 (d) 2477.57 J mol −1

16. C6L7V6Q8 The standard enthalpies of formation of CO 2 (g), H 2 O(l) and glucose(s) at 25°C are −400 kJ mol −1 , −300 kJ mol −1 and −1300 kJ mol −1 , respectively. The standard enthalpy of combustion per gram of glucose at 25°C is: (a) +2900 kJ (b) −2900 kJ (c) −16.11 kJ (d) +16.11 kJ

17. C6L8VSuggQ2 H 2 (g) + (1/2)O 2 (g) → H 2 O(l); ΔH = −68 kcal. The heat change for the decomposition of 3.6 g of water is: (a) 136 kcal (b) 13.6 kcal (c) 1.36 kcal (d) 68 kcal

18. C6L10V5Q6 What is the change in entropy when 2.5 mole of water is heated from 27°C to 87°C? Assume that the heat capacity is constant. (C p,m (H 2 O) = 4.2 J/g-K, ln(1.2) = 0.18) (a) 16.6 J/K (b) 9 J/K (c) 34.02 J/K (d) 1.89 J/K

19. C6SBVPYQQ90 At 25°C and 1 atm pressure, the enthalpies of combustion are: H 2 = −286.0, C(graphite) = −394.0, C 2 H 6 (g) = −1560.0 kJ mol −1 . The enthalpy of formation of ethane is: (a) +54.0 kJ mol −1 (b) −68.0 kJ mol −1 (c) −86.0 kJ mol −1 (d) +97.0 kJ mol −1

Substance	H_2	C (graphite)	C_2H_6 g
$\frac{\Delta_c H^\ominus}{kJmol^{-1}}$	-286.0	-394.0	-1560.0

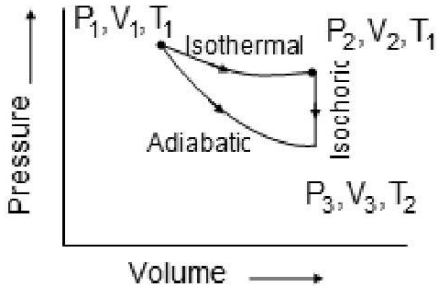
PART B · Q21–25 · NUMERICAL VALUE

21. C2XVRT1Q46 In the hydrogen spectrum, the longest wavelength in the Lyman series is 120 nm and the shortest wavelength in the Balmer series is 360 nm. From this data, the longest wavelength (in nm) of light that could ionize a hydrogen atom is ___.

22. C2XVRT3Q47 If the de-Broglie wavelength of the fourth Bohr orbit of hydrogen atom is 4 Å, the circumference (in Å) of the orbit will be :

20. C6XVCAT6Q30 An ideal diatomic gas is caused to pass through a cycle shown on the P-V diagram in figure, where $V_2 = 3.00 V_1$. If P_1, V_1 and T_1 specify the state 1, then the temperature of the state 3 is (a) $(T_1/3)^1$ (b) $(T_1/3)^4$ (c) $T_1(1/3)^{(0.4)}$ (d) $(T_1/3)^2$

3 is



23. C3XVRT2Q48 The chloride of a metal contains 71% chlorine by mass and the vapour density of it is 50. The atomic mass of the metal will be:

24. C4XVRT3Q49 If for an element (X), the values of successive ionisation energies I 1 , I 2 , I 3 , I 4 and I 5 are 800, 2427, 3658, 25024, 32824 kJ mol −1 respectively, then the number of valence electrons present are :

25. C5SBVPYQQ41 The total number of anti bonding molecular orbitals, formed from 2s and 2p atomic orbitals in a diatomic molecule is _____

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M4L1V6Q7 If p, q, r are any real numbers, then (a) max(p, q) < max(p, q, r) (b) min(p, q) = (1/2)(p + q − |p − q|) (c) max(p, q) < min(p, q, r) (d) None of these

2. M4L3V4Q5 Solve for the sum of all real roots of |x − 1| + sgn(x − 1) = 2. (a) −2 (b) 2 (c) 0 (d) 4

3. M5ASBVPYQQ4 If tan A = $1/\sqrt{(x(x^2 + x + 1))}$, tan B = $\sqrt{x}/\sqrt{(x^2 + x + 1)}$ and tan C = $(x^{(-3)} + x^{(-2)} + x^{(-1)})^{(1/2)}$, 0 < A, B, C < π/2, then A + B is equal to : (a) C (b) π − C (c) 2π − C (d) π/2 − C

4. M5CSBV8Q3 If x = sin(α−β)·sin(γ−δ), y = sin(β−γ)·sin(α−δ), z = sin(γ−α)·sin(β−δ), then : (one or more options may be correct) (a) x + y + z = 0 (b) $x^3 + y^3 + z^3 = 3xyz$ (c) x + y − z = 0 (d) $x^3 + y^3 - z^3 = 3xyz$

5. M6L2V5Q6 If (x − 2)/(x² + 1) < −1/2, then 'x' belongs to: (a) (1, 3) (b) (−1, 3) (c) (−3, 1) (d) (−3, −1)

6. M7L4V6Q7 The set of values of x satisfying the inequality $1/\log_4((x+1)/(x+2)) \le 1/\log_4(x+3)$ is: (a) (−3, −2) (b) (−3, −2) ∪ (−1, ∞) (c) (−1, ∞) (d) none of these

7. M7SBVPYQQ4 The product of all solutions of the equation $e^{(5(\log_e x)^2+3)}=x^8$, x>0, is: (a) $e^{(8/5)}$ (b) $e^{(6/5)}$ (c) e (d) e^2

8. M7XVCAT4Q65 If the equation $\log_{12}[\log_8(\log_4 x)] / \log_5[\log_4(\log_y(\log_2 x))] = 0$ has a solution for x when c < y < b, y ≠ a, where b is as large as possible and c is as small as possible, then the value of (a + b + c) is equal to: (a) 18 (b) 19 (c) 20 (d) 21

9. M8AL1V6Q2 If the m th terms of the series 63 + 65 + 67 + 69 + ... and 3 + 10 + 17 + 24 + ... are equal, then m = (a) 11 (b) 12 (c) 13 (d) 15

10. M8BL1V6Q6 If x > 1, y > 1, z > 1 are in G.P., then log_(ex) e, log_(ey) e, log_(ez) e are in: (a) A.P. (b) H.P. (c) G.P. (d) A.G.P.

11. M8BL3V4Q4 The 20 th term of the series 2×4 + 4×6 + 6×8 + ... will be (a) 1600 (b) 1680 (c) 420 (d) 840

12. M8CL1V2Q5 Find the 3 rd term of the H.P. corresponding to the A.P. 1, 2, 3, ... (a) 1/2 (b) 1/3 (c) 3 (d) 2

13. M8DL1V4Q3 If T n = 2n² + 5n, find S n . (a) n(n+1)(n+2)/6 (b) n(n+1)(4n − 17)/6 (c) n(n+1)(4n + 17)/6 (d) None of these

14. M8DL5V4Q6 The Geometric mean of 1, 2, 2², ..., 2^n is (a) $2^{(n/2)}$ (b) $n^{((n+1)/2)}$ (c) $2^{(n(n+1)/2)}$ (d) $2^{((n-1)/2)}$

15. M9L2V1Q4 What is the multiplicative inverse of a complex number z? (a) z (b) -z (c) 1/z (d) z^2

16. M9L2V6Q1 Consider the following expressions involving complex numbers: $E_1 = (3 - i)/(2 + i) + (3 + i)/(2 - i)$, $E_2 = 3/(1 + i) - 2/(2 - i) + 2/(1 - i)$. Which of the following statements are correct? (a) E_1 is a pure real number. (b) The real part of E_2 is positive. (c) (3 - i)/(2 + i) = 1 - i (d) (3 + i)/(2 - i) = 1 + i (one or more options may be correct)

17. M9L2VSuggQ1 What is the modulus of a complex number z = x + iy? (a) x + y (b) $\sqrt{(x^2 + y^2)}$ (c) $x^2 - y^2$ (d) x + iy

18. M9L4V5Q7 If z_1 and z_2 are any two complex numbers, then $|z_1 + z_2|^2 + |z_1 - z_2|^2$ is equal to (a) $2|z_1|^2|z_2|^2$ (b) $2|z_1|^2 + 2|z_2|^2$ (c) $|z_1|^2 + |z_2|^2$ (d) $2|z_1||z_2|$

19. M9L5VSuggQ13 Let $z = (\cos\theta + i\sin\theta)/(\cos\theta - i\sin\theta)$, $\pi/4 < \theta < \pi/2$. Then principle value of amp(z) is : (a) 2θ (b) $2\theta - \pi$ (c) $\pi + 2\theta$ (d) None of these

PART B · Q21–25 · NUMERICAL VALUE

21. M2XVRT2Q62 Out of 800 boys in a school, 224 played cricket, 240 played hockey and 336 played basketball. Of the total, 64 played both basketball and hockey; 80 played cricket and basketball and 40 played cricket and hockey; 24 played all the three games. The number of boys who did not play any game is: (A) 128 (B) 216 (C) 240 (D) 160

22. M3XVCAT3Q71 Find value of $\cos 570° \sin 510° + \sin(-330°) \cos(-390°)$

20. M9L7V3Q3 If ω is one of the complex cube roots of unity then $(1 - \omega + \omega^2)(1 + \omega - \omega^2) =$ (a) 0 (b) 4 (c) 3ω (d) 1

23. M6XVRT3Q71 The minimum value of the function $f(x) = 2|x - 1| + |x - 2|$ is

24. M8AXVRT4Q73 The number of common terms in the following sequences (each of which is an A.P) 37, 103, 169, ... and 17, 82, 147, ... which are less than 10,000 is

25. M8DXVCAT6Q74 Sum series $S = 1 + 3/2 + 5/2^2 + 7/2^3 + \ldots \infty$ is